Bachelor of Technology
in Electrical Engineering

Curriculum and Syllabi

DEPARTMENT OF ELECTRICAL ENGINEERING
Z. H. COLLEGE OF ENGINEERING AND TECHNOLOGY
ALIGARH MUSLIM UNIVERSITY, ALIGARH
2021-22

About the Department

The department of Electrical Engineering was established in 1935 with an intake of 20 students, and the first graduate program started in 1940. A new phase of significant developments in the department started during the third five-year plan. Currently, the department is offering undergraduate, postgraduate, and doctorate programs in various specialized disciplines of electrical engineering. Presently the total intake of the B.Tech. program is 60, and the M. Tech program is 54. The department recently started a new M.Tech. program in "Solar and Renewable Energy," with an intake of 10 students. The department now runs one B.Tech and four M.Tech. Programs. The thrust areas of the department are Power Systems, Power Electronics, Control, and High Voltage Engineering.

The faculty members have published a significant number of research articles in repueted journals and have also published several textbooks/edited books with reputed international publishers. Many research projects sponsored by A.I.C.T.E, U.G.C., D.S.T., and C.S.T. (U.P.) have been successfully completed. The first PhD in the Faculty of Engineering & Technology, Aligarh Muslim University, was produced by this department in 1977. Since then, many students have completed their PhD degrees in the department. Further, the department's faculty members have published more than 1000 research papers. The department organized one IEEE international conference in 2019. Many Workshops/Seminars/Training programs have been successfully organized in the last five years. Faculty members have chaired technical sessions of different national and international meets and have been regular members of various technical committees.

The department of Electrical Engineering has sanctioned strength of 27 faculty members, and most faculty members hold doctorates from reputed national and international institutes. Faculty members have specialization in diverse fields of Electrical Engineering such as Power systems, Electrical Machines, Electrical Drives, High Voltage Engineering, Control systems, Instrumentation, Biomedical Engineering, Hybrid Electric Vehicles, and Renewable sources of Energy.

VISION OF THE DEPARTMENT

To become an internationally acclaimed department for the advancement of knowledge and technologies in the field of Electrical Engineering.

MISSION OF THE DEPARTMENT

1. To become an internationally leading department for higher learning.
2. To build upon the culture and values of universal science and contemporary education.
3. To be a center of research and education generating knowledge and technologies which lay ground work in shaping the future in the field of Electrical Engineering.
4. To develop collaboration with industries, R&D organizations, and government agencies and actively participate in conferences, and technical and community activities.

PROGRAM SPECIFIC OUTCOMES (PSOs)

B. TECH. (ELECTRICAL ENGINEERING) PROGRAM

PSO1 Graduates will be able to analyze the concepts of generation, transmission, distribution, and measurement of electrical power and its conversion and control.

PSO2 Graduates will be able to design and develop systems in the emerging electrical and allied disciplines to meet out the industry challenges.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1 Students will have a successful technical and professional careers, including supportive and leadership roles on multidisciplinary teams.

PEO 2 Students will be able to acquire, use and develop skills as required for effective professional practices.

PEO 3 Students will be able to attain holistic education that is an essential prerequisite for being a responsible member of society.

PEO 4 Students will be engaged in life-long learning, to remain abreast in their profession and be leaders in our technologically vibrant society.

PROGRAM OUTCOMES (POs)

B. TECH. (ELECTRICAL ENGINEERING) PROGRAM

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Lifelong learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Course Structure

B.Tech. in Electrical Engineering

Z.H. College of Engineering Technology, Aligarh Muslim University Aligarh

Course Structure: B.Tech.(Electrical Engineering)
Valid for the students admitted from year 2017 Onwards

Semester 1: Sections A, B, C

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	BS	AMS1110	Applied Mathematics-I	3	1	0	4	15	25	60	100
2	BS	ACS1110	Applied Chemistry	3	1	0	4	15	25	60	100
3	ESA	EEA1110	Principles of Electrical Engineering	2	1	0	3	15	25	60	100
4	ESA	CEA1110	Environmental Studies	2	1	0	3	15	25	60	100
5	ESA	MEA1110	Engineering Thermodynamics	3	1	0	4	15	25	60	100
6	BS	ACS1910	Applied Chemistry Lab	0	0	3	1.5	60		40	100
7	ESA	COA1910	Computer Programming Lab	0	0	3	1.5	60		40	100
8	ESA	MEA1910	Engineering Graphics Lab	0	1	1	2	60		40	100

TOTAL CREDITS: 23

Semester 2: Sections A, B, C

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	BS	AMS1120	Applied Mathematics-II	3	1	0	4	15	25	60	100
2	BS	APS1110	Applied Physics	3	1	0	4	15	25	60	100
3	ESA	ELA1110	Principles of Electronics Engineering	2	1	0	3	15	25	60	100
4	ESA	CEA1120	Strength of Materials	2	1	0	3	15	25	60	100
5	ESA	MEA1120	Engineering Mechanics	2	1	0	3	15	25	60	100
6	HM	EZH1110	English	2	1	0	3	15	25	60	100
7	BS	APS1910	Applied Physics Lab	0	0	3	1.5	60		40	100
8	ESA	MEA1920	Manufacturing Process Lab	0	0	3	1.5	60		40	100

TOTAL CREDITS: 23

Z.H. College of Engineering Technology, Aligarh Muslim University Aligarh

Course Structure: B.Tech.(Electrical Engineering)
Valid for the students admitted from year 2017 Onwards

Semester 1: Sections D, E, F

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	BS	AMS1110	Applied Mathematics-I	3	1	0	4	15	25	60	100
2	BS	APS1110	Applied Physics	3	1	0	4	15	25	60	100
3	ESA	ELA1110	Principles of Electronics Engineering	2	1	0	3	15	25	60	100
4	ESA	CEA1120	Strength of Materials	2	1	0	3	15	25	60	100
5	ESA	MEA1120	Engineering Mechanics	2	1	0	3	15	25	60	100
6	HM	EZH1110	English	2	1	0	3	15	25	60	100
7	BS	APS1910	Applied Physics Lab	0	0	3	1.5	60		40	100
8	ESA	MEA1920	Manufacturing Process Lab	0	0	3	1.5	60		40	100

TOTAL CREDITS: 23

Semester 2: Sections D, E, F

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	BS	ACS1110	Applied Chemistry	3	1	0	4	15	25	60	100
2	ESA	EEA1110	Principles of Electrical Engineering	2	1	0	3	15	25	60	100
3	ESA	CEA1110	Environmental Studies	2	1	0	3	15	25	60	100
4	ESA	MEA1110	Engineering Thermodynamics	3	1	0	4	15	25	60	100
5	BS	ACS1910	Applied Chemistry Lab	0	0	3	1.5	60		40	100
6	ESA	COA1910	Computer Programming Lab	0	0	3	1.5	60		40	100
7	ESA	MEA1910	Engineering Graphics Lab	0	1	1	2	60		40	100
8	BS	AMS1120	Applied Mathematics-II	3	1	0	4	15	25	60	100

TOTAL CREDITS: 23

Course Structure: B.Tech.(Electrical Engineering)
Valid for the students admitted from year 2017 Onwards

Semester 3:

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	BS	AMS2230	Higher Mathematics	3	1	0	4	15	25	60	100
2	BS	APS2050	Electrical Engineering Materials	2	1	0	3	15	25	60	100
3	DC	EEC2110	Electrical Machines-I	3	1	0	4	15	25	60	100
4	DC	EEC2710	Circuit Theory	3	1	0	4	15	25	60	100
5	DC	EEC2720	Electromagnetic Field Theory	3	1	0	4	15	25	60	100
6	DC	EEC2730	Signals & Systems	2	1	0	3	15	25	60	100
7	HM	EZH2910	Communication Skills Lab	0	1	2	2	60	0	40	100
8	DC	EEC2910	Electrical Machines Lab I	0	1	2	2	60	0	40	100

TOTAL CREDITS: 26

Semester 4:

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	ESA	ELA2010	Logic & Digital Circuit	3	1	0	4	15	25	60	100
2	HM	MEH2450	Engineering Economy & Management	3	1	0	4	15	25	60	100
3	DC	EEC2120	Electrical Machines II	3	1	0	4	15	25	60	100
4	DC	EEC2210	Power Electronics-I	3	1	0	4	15	25	60	100
5	DC	EEC2310	Power System Engineering	3	1	0	4	15	25	60	100
6	DC	EEC2510	Electrical Measurement	3	1	0	4	15	25	60	100
7	DC	EEC2920	Electrical Machines Lab II	0	1	2	2	60	0	40	100
8	DC	EEC2930	Circuits and Measurements Lab	0	1	2	2	60	0	40	100

TOTAL CREDITS: 28

Course Structure: B.Tech.(Electrical Engineering)
Valid for the students admitted from year 2017 Onwards

Semester 5:

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	ESA	ELA3020	Fundamentals of Communication Engineerin	3	1	0	4	15	25	60	100
2	DC	EEC3210	Power Electronics-II	3	1	0	4	15	25	60	100
3	DC	EEC3310	Power System Analysis	3	1	0	4	15	25	60	100
4	DC	EEC3510	Electrical & Electronic Instr.	2	1	0	3	15	25	60	100
5	DC	EEC3610	High Voltage Engineering	2	1	0	3	15	25	60	100
6	ESA	ELA3910	Electronics Engg. Lab	0	1	2	2	60	0	40	100
7	DC	EEC3910	Power Electronics Lab	0	1	2	2	60	0	40	100
8	OE	---------	Open Elective-1	3	1	0	4	15	25	60	100

TOTAL CREDITS: 26

Semester 6:

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	DC	EEC3110	Electrical Drives	3	1	0	4	15	25	60	100
2	DC	EEC3220	New and Renewable Energy Sources	3	1	0	4	15	25	60	100
3	DC	EEC3320	Electrical Power Gen. & Utilization	3	1	0	4	15	25	60	100
4	DC	EEC3410	Dynamic System Analysis	3	1	0	4	15	25	60	100
5	DC	EEC3710	Microcontroller Systems and Appl.	3	1	0	4	15	25	60	100
6	DC	EEC3920	Power System and High Voltage Lab	0	1	2	2	60	0	40	100
7	DC	EEC3930	Instrumentation Lab	0	1	2	2	60	0	40	100
8	HM	---------	Elective Course under Humanities category	3	1	0	3	15	25	60	100

TOTAL CREDITS: 27

Course Structure: B.Tech.(Electrical Engineering)
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Semester 7:

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	DC	EEC4310	Power System Protection	3	1	0	4	15	25	60	100
2	DC	EEC4410	Control Systems	3	1	0	4	15	25	60	100
3	DC	EEC4910	Power System Protection Lab	0	1	2	2	60	0	40	100
4	DC	EEC4920	Control Lab	0	1	2	2	60	0	40	100
5	DC	EEC4930	Electric Machine Design	0	1	2	2	60	0	40	100
6	DC	EEC4940	Power System Design	0	1	2	2	60	0	40	100
7	DC	EEC4980	Project Phase-I	0	2	0	2	60	0	40	100
8	DE	---------	Departmental Elective-1	3	1	0	3	15	25	60	100
9	DE	---------	Departmental Elective-2	3	1	0	3	15	25	60	100
10	DE	---------	Departmental Elective-3	3	1	0	3	15	25	60	100

TOTAL CREDITS: 27

Semester 8:

S.No.	Couse Catag.	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	DC	EEC4990	Project Phase-II	0	4	0	4	60	0	40	100
1	DE	---------	Departmental Elective-4	3	1	0	3	15	25	60	100
2	DE	---------	Departmental Elective-5	3	1	0	3	15	25	60	100
3	DE	---------	Departmental Elective-6	3	1	0	3	15	25	60	100
4	DE	---------	Departmental Elective-7	3	1	0	3	15	25	60	100
5	OE	---------	Open Elective-2	3	1	0	4	15	25	60	100

TOTAL CREDITS: 20

Course Structure: B.Tech.(Electrical Engineering)
Valid for the students admitted from year 2017 Onwards

List of Departmental Elective (DE) Courses:

S.No.	Crs. No.	Course title	S.No.	Crs. No.	Course title
1	EEE4310	Power Station Practice	2	EEE4210	Power Semiconductor Controllers
3	EEE4110	Modern Electric Drives	4	EEE4230	Solar Photovoltaic Systems
5	EEE4320	Power Quality	6	EEE4220	Modelling and Control of Electronics converters
7	EEE4120	Electric Vehicles & Traction	8	EEE4420	Embedded Systems
9	EEE4710	Numerical Computational Methods in Electrical Engg.	10	EEE4350	HVDC and FACTS Technology
11	EEE4720	Machine Learning	12	EEE4520	Process Instrumentation & Control
13	EEE4740	Digital Signal Processing	14	EEE4360	Smart Grid
15	EEE4130	Generalized Theory of Electrical Machines	16	EEE4330	Digital Simulation of Power System
17	EEE4340	Power System Reliability	18	EEE4440	Mechatronics
19	EEE4430	Nonlinear Control System	20	EEE4730	Artificial Neural Networks
21	EEE4510	Measurement and Data Acquisition	22		

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